

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

Short-Term Rentals and Overtourism: A Data Science Perspective on Barcelona

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Abstract

The proliferation of short-term rental platforms such as Airbnb has generated measurable tensions between tourism economies and residential housing markets in major European cities. This study analyzes the socio-economic impact of short-term rentals on housing accessibility and resident displacement in Barcelona, Spain, using a longitudinal dataset spanning March 2021 to February 2026. The dataset comprises 894 unique listings across 11 neighborhoods, analyzed through spatio-temporal methods, K-Means clustering, and a Random Forest predictive model. Results indicate that 63% of active listings consist of entire residential units, that average monthly STR revenue reaches €2,210, and that 73% of Barcelona’s neighborhoods exhibit high vulnerability to STR-driven displacement according to a composite STR Pressure Index. Eixample and Gràcia are identified as the most critically affected neighborhoods. The predictive model adopts a time-lagged framework in which next month’s neighborhood occupancy rate is predicted from current month market characteristics, achieving an R^2 of 0.732 on the held-out test set and a cross-validated R^2 of 0.682, with availability ratio and current occupancy emerging as the dominant predictors. Regulatory scenario simulations indicate that primary residence requirements produce the smallest increase in predicted occupancy among remaining operators (+2.0% above baseline), making them the most targeted intervention among those tested, while 30-day annual operating caps risk concentrating tourist demand into fewer available days, potentially intensifying rather than reducing the commercial character of surviving STR operations. The study concludes with evidence-based policy recommendations addressing the balance between tourism economic benefits and residents’ right to housing.

1 Introduction

Barcelona is a useful case for studying how short-term rentals (STRs) reshape urban housing markets. The city combines strong tourism demand, limited central housing supply, and growing public concern about overtourism. This study analyzes Airbnb activity in Barcelona from March 2021 to February 2026, covering 894 unique listings across 11 neighborhoods. Over this period, active listings increased from about 210 to a peak of about 540, showing a strong post-pandemic recovery despite protests and regulatory debate.

The central problem is that STRs can remove dwellings from the long-term rental market, especially when entire homes are operated as commercial tourist accommodation. This conversion pressure matters in Barcelona because rents have risen sharply and the city has announced a plan to phase out tourist apartment licenses by 2028. The report therefore evaluates where STR pressure is most concentrated, which market features are associated with displacement risk, and which regulatory scenarios appear most effective.

The analysis is guided by four research questions:

- How has STR growth affected housing availability and displacement pressure?
- Which neighborhoods show the highest vulnerability to STR-driven displacement?
- How does Barcelona compare with other European cities facing overtourism?
- Which regulatory measures best reduce risk while preserving tourism benefits?

2 Problem Description

Barcelona's STR market has moved beyond occasional home-sharing toward a more commercial model. In the cleaned dataset, 63% of active listing-months are entire homes or apartments, while shared rooms represent only 0.7%. Host activity is also concentrated: 36.6% of hosts operate more than one property, and medium-portfolio hosts account for 8% of active listing-months. These patterns suggest that a substantial share of STR supply competes with residential housing rather than simply monetizing spare capacity.

STRs affect housing through three main channels. First, dwellings converted to tourist use are removed from the long-term rental stock. Second, high STR revenues increase the opportunity cost of renting to residents, encouraging further conversion. Third, spatial concentration in central and high-demand neighborhoods intensifies local displacement pressure. In this study, Eixample, Gràcia, les Corts, and Sarrià - Sant Gervasi are treated as key pressure areas because they combine high prices, strong demand, and substantial STR activity.

The consequences extend beyond individual rent burdens. Resident displacement can weaken neighborhood networks, reduce access to central employment and services, and accelerate commercial change toward tourist-oriented businesses. Barcelona therefore faces a policy trade-off: tourism generates income and employment, but unregulated STR growth can undermine housing affordability and residential stability.

The report operationalizes this problem through three empirical tasks: mapping STR intensity, constructing neighborhood-level pressure and displacement indicators, and simulating policy interventions such as license reductions, day caps, and primary-residence requirements.

3 Literature Review

3.1 Global Evidence on Short-Term Rentals and Housing Markets

Research on the relationship between short-term rental platforms and urban housing markets has expanded substantially since 2015. The foundational empirical contribution is provided by Barron, Kung, and Proserpio (2021) [4], whose instrumental variable analysis of US-wide data established a causal relationship between Airbnb market penetration and housing costs, finding that a 1% increase in listings produces a 0.018% increase in rents and a 0.026% increase in house prices. Effects are concentrated in areas with low owner-occupancy rates, consistent with theoretical predictions about the conditions under which STR conversion incentives are strongest.

Wachsmuth and Weisler (2018) [22] provide the dominant theoretical framework for understanding STR-driven displacement through their application of rent gap theory to platform economics. The rent gap model holds that property owners convert residential units to STRs when potential STR ground rent exceeds capitalized residential ground rent by a margin sufficient to justify conversion costs. This framework explains the spatial clustering of STR activity in tourism-attractive areas and the tendency for displacement pressure to intensify as platform penetration increases and neighborhood composition shifts. The model predicts that displacement pressures will be strongest not necessarily in the most expensive central locations but in neighborhoods where the gap between potential STR revenues and existing residential rents is largest — a prediction our revenue analysis confirms in the Barcelona context, where Gràcia shows the highest average monthly revenue at €3,021 despite not being the neighborhood with the highest listing density.

Celata and Romano (2022) [7] document the overtourism phenomenon in Italian cities, finding that STR nightly earnings in central neighborhoods frequently reach three to four times monthly conventional rental income, establishing the financial basis for large-scale residential-to-tourist conversion. Gurran and Phibbs (2017) [14] provide evidence on the neighborhood-level externalities generated by STR concentrations, documenting noise, security, and community cohesion effects that contribute to long-term resident departure independently of direct rental price effects.

3.2 Case Studies: Milan, Lisbon, Amsterdam, and New York

Milan provides the most structurally comparable European case to Barcelona, sharing characteristics as a major metropolitan center facing significant tourism pressure alongside constrained housing supply. Analysis of Milan’s STR market documents that the Navigli district experienced 42% rent growth between 2015 and 2023 compared to a 28% citywide average, consistent with the spatial concentration of STR-gentrification effects predicted by theoretical models. Multi-property hosts control approximately 40% of all Airbnb listings in Milan, a commercialization pattern closely paralleling our finding that 36.6% of Barcelona listing-months are controlled by multi-property operators.

Lisbon offers an instructive parallel given its comparable position as a Southern European city experiencing intense tourism pressure. Research by Cocola-Gant and Gago (2021) [9] documents that Airbnb expansion produced 37% rental price growth between 2015 and 2019, accompanied by collective displacement in which entire residential communities dispersed as buildings were systematically converted to tourist accommodation. The Lisbon case illustrates a critical policy lesson: regulatory measures introduced after substantial market penetration has already occurred proved insufficient to reverse the touristification processes already underway in the city’s historic neighborhoods.

Amsterdam’s experience provides evidence on the relationship between regulatory intervention and housing affordability outcomes that directly informs our scenario analysis. The city implemented progressive annual letting limits — reduced from 60 to 30 days in 2018 — producing a 54% reduction in active STR listings between 2019 and 2024. However, this regulatory success did not translate into measurable housing affordability improvement, with rental prices continuing to rise and social housing waiting lists reaching 13 years according to Bei and Celata (2023) [6]. The Amsterdam case challenges

the assumption that reducing STR supply automatically produces housing accessibility benefits, a finding our scenario analysis corroborates from a different angle: our simulated 30-day cap scenario produces a counterintuitive 78.1% increase in predicted occupancy among remaining operators, indicating that day-cap restrictions risk concentrating tourist demand into fewer available days rather than reducing it. This scarcity-driven demand concentration suggests that supply restrictions which cut listed availability may intensify, rather than relieve, the commercial pressure on the surviving STR stock — consistent with Amsterdam’s experience of falling listing counts alongside persistent housing market strain.

Research on New York City by Wachsmuth et al. (2018) documents that the city’s strict host-presence requirement produced an 80% reduction in Airbnb listings but primarily displaced operators to adjacent jurisdictions rather than returning housing to the long-term rental market. London’s 90-day annual letting limit showed similar limitations, with Shabrina, Arcaute, and Batty (2021) [20] finding that 35% of entire-home listings exceeded the regulatory limit due to inadequate enforcement capacity.

3.3 Gaps and Controversies in Existing Research

Significant methodological limitations persist in the STR literature. The most fundamental challenge is causal identification: most available evidence is observational, making it difficult to isolate the effect of STR platforms from simultaneous gentrification, tourism development, and broader housing market pressures affecting the same neighborhoods. Instrumental variable approaches address this challenge partially, but the external validity of findings derived from US housing market contexts to European markets with different regulatory traditions and tenure structures remains uncertain.

The distributional consequences of STR market growth across income groups, age cohorts, and household types have received insufficient research attention. Aggregate evidence documents neighborhood-level rental price effects but provides limited insight into how these effects are distributed within affected populations. Available evidence suggests disproportionate impacts on renters, younger households, and lower-income families, but rigorous quantification of these distributional effects is lacking.

The small number of observations available at the neighborhood level in many studies — including the present one, which analyzes 11 neighborhoods — limits the statistical power of correlation analyses and constrains the precision of vulnerability assessments. Our study addresses this limitation through longitudinal monthly analysis covering 60 months, but the constraint on neighborhood-level statistical power remains a genuine limitation that we address explicitly in the methodological discussion.

4 Indicators

4.1 STR Listing Density per Neighborhood

STR listing density is defined as the number of unique active listings per neighborhood per month, computed from the `listing_id` and `neighborhood` fields of the study dataset aggregated at monthly intervals. This indicator captures the basic scale of platform market penetration in each area and provides the spatial foundation for vulnerability assessment. The indicator reveals substantial geographic concentration: Eixample records 308 unique listings while Nou Barris records 12, a ratio of approximately 26:1 across neighborhoods within the same city. This concentration is consistent with the theoretical prediction that STR activity clusters in tourism-attractive central areas and provides an empirical basis for spatially differentiated regulatory responses.

Listing density is subject to measurement limitations that must be acknowledged. The dataset captures only listings active on a single platform during the data collection period and excludes properties listed on competing platforms or through informal channels. Neighborhood classifications correspond to Barcelona’s major administrative districts rather than finer-grained spatial units, meaning within-

district variation in market intensity cannot be captured. Density estimates therefore represent lower bounds on actual STR market penetration at the neighborhood level.

4.2 Housing Availability Ratio

The housing availability ratio is computed as the proportion of available days relative to total calendar days for listings within each neighborhood per month: available days divided by the sum of available days and unavailable days. A higher ratio indicates a larger proportion of time during which properties are listed as available for tourist booking, functioning as a proxy for the degree to which residential housing stock has been dedicated to STR use. The indicator ranges from 24.5% in Districte II to 63.7% in Nou Barris across the study neighborhoods.

The interpretation of this indicator requires contextual qualification. Low availability ratios in high-demand tourist neighborhoods reflect continuous tourist occupation rather than residential use, while high availability ratios in lower-demand areas may reflect properties listed but infrequently booked. The indicator must therefore be interpreted in conjunction with occupancy and revenue data to accurately characterize the nature and intensity of STR market activity in each neighborhood.

4.3 Occupancy Rate as a Proxy for Commercial Intensity

Occupancy rate is defined as the proportion of unavailable days — days on which a listing is booked — relative to total calendar days. This indicator provides the most direct available measure of how intensively listed properties are used for tourist accommodation. An occupancy rate exceeding 50% is applied as a threshold indicator of commercial operation, consistent with the operational intensity characteristic of professional accommodation providers rather than occasional home-sharers. Four neighborhoods in the dataset exceed this threshold: Districte II at 75.6%, Gràcia at 57.2%, Horta-Guinardó at 55.6%, and Eixample at 54.5%.

Seasonality analysis reveals that occupancy rates fluctuate substantially across calendar months, with July recording the highest average revenue at €2,727 per listing and January the lowest at €1,513. This seasonal variation generates concentrated tourist pressure during summer months, corresponding temporally to the period of the July 2024 protests and confirming the relationship between peak STR market activity and the social tensions that erupted in that period.

4.4 STR Revenue as a Displacement Pressure Indicator

Monthly revenue per listing, derived from the dataset’s revenue field representing estimated monthly earnings per active listing, provides the most direct measure of the financial incentive driving residential-to-tourist conversion. This indicator ranges from an average of €395 per month in Nou Barris to €3,021 in Gràcia. The revenue differential between neighborhoods reflects both nightly pricing differences and booking intensity variation, and constitutes the primary economic driver of STR market concentration in high-demand areas.

While revenue quantifies the economic incentive for conversion, its analytical role must be contextualized. Rather than serving as the dominant predictor of short-term occupancy fluctuations (where market momentum indicators like availability ratio prevail, as detailed in Section 6.6), revenue functions as a foundational metric for evaluating long-term spatial vulnerability. By identifying the specific locations where the rent gap between potential tourist income and conventional residential rent is widest, revenue provides the necessary economic context for interpreting our composite vulnerability indices and the subsequent neighborhood clustering.

4.5 Ethical and Social Implications of the Selected Indicators

The indicators selected for this analysis are derived entirely from supply-side platform data, capturing the behavior of hosts and platform operators but not the direct experiences of residents affected by

STR-driven displacement. This data asymmetry is a structural limitation of platform-based research: the populations most adversely affected by STR market growth — displaced renters, communities experiencing social disruption — are not represented in the platform data from which our indicators are constructed. The indicators measure market conditions associated with displacement risk rather than displacement itself and should be interpreted accordingly.

The construction of composite indicators such as the STR Pressure Index involves normalization and weighting choices that embed analytical assumptions about the relative importance of different market characteristics. While our methodology is transparent and reproducible, the composite scores represent one operationalization among several possible alternatives. Neighborhoods with high revenue but low listing density may face intense conversion pressure that aggregate density-weighted indices understate. Policy applications of vulnerability scores should therefore be supplemented by qualitative assessment of local housing market conditions rather than applied mechanically.

The use of data science methods to identify displacement risk also raises questions about the ethical use of predictive analytics in housing policy contexts. Predictive models trained on market data may reproduce or amplify existing inequalities if their outputs are used to prioritize regulatory resources in ways that systematically disadvantage already-vulnerable communities. Policy makers applying our vulnerability scores should ensure that the populations most directly affected by STR market dynamics are meaningfully involved in the policy processes informed by these analyses.

5 Data Collection and Preparation

The analysis uses a monthly Airbnb dataset, downloaded from Kaggle [1], for Barcelona from March 2021 to February 2026, supplemented by neighborhood-level aggregation and engineered indicators. The raw files include listing identifiers, host identifiers, room type, neighborhood, active status, availability, occupancy, average daily rate, revenue, RevPAR, booking lead time, and length of stay. After cleaning, the analytical dataset contains 24,467 active listing-months, 894 unique listings, and 684 unique hosts across 11 neighborhoods.

Preprocessing was carried out in Python and included five main steps. First, malformed UTF-8 neighborhood names were corrected and standardized. Second, monthly date fields were parsed to enable time-series aggregation. Third, observations with missing neighborhood values were removed, while booking-related missing values were retained where they reflected months without confirmed bookings. Fourth, inactive listing-months were excluded. Fifth, extreme outliers in average daily rate, revenue, and RevPAR were removed using the interquartile range rule, reducing distortion from implausibly high values.

Several variables were then engineered for analysis. Hosts were classified as single-listing, small-portfolio, or medium-portfolio operators based on the number of unique listings attached to each host ID. Housing availability was measured as available days divided by total calendar days. The STR Pressure Index was computed at the neighborhood-month level using normalized listing count, occupancy, average daily rate, entire-home ratio, and professional-host ratio. A separate displacement risk score, combining normalized occupancy, revenue, entire-home ratio, professional-host ratio, and the inverse availability ratio, was constructed as a descriptive vulnerability indicator. This composite score was not used as the target variable in predictive modeling: as detailed in Section 6.6.1, the model instead predicts next-month neighborhood occupancy to avoid the circularity of training on a composite of its own features.

The main limitations are that the dataset covers Airbnb rather than the full STR market, lacks direct long-term rent data, and provides only 11 neighborhoods for spatial comparison. Revenue is also estimated from platform data rather than observed transaction records. These limits mean the results should be interpreted as evidence of STR pressure and relative vulnerability, not as a complete causal estimate of rent inflation.

6 Data Analysis and Visualization

6.1 Spatio-Temporal Analysis of STR Distribution in Barcelona

The cleaned analytical dataset comprises 24,467 active listing-months representing 894 unique listings across 11 neighborhoods, covering March 2021 through February 2026. The STR market in Barcelona demonstrates substantial spatial concentration and clear temporal growth patterns that together characterize the nature and intensity of platform penetration across the city’s districts.



Figure 1: Active Airbnb Listings Over Time in Barcelona (2021–2026)

Figure 1 presents the monthly evolution of active unique listings across the study period. The market grew from 210 active listings in March 2021 to a peak of approximately 530 by mid-2022, reflecting the rapid recovery of tourism demand following pandemic-related restrictions. The market maintained elevated listing counts through 2023 before contracting notably in early 2024, reaching approximately 300 listings by mid-2024. A subsequent recovery brought the market back to peak levels of approximately 540 listings by mid-2025, after which a moderate decline is observed through February 2026. The July 2024 protests, marked by the vertical dashed line, coincide with the post-contraction recovery phase rather than with any sustained market decline, indicating that social pressure did not produce a durable reduction in STR market activity. This is consistent with the broader European evidence — including the Amsterdam case documented by Bei and Celata (2023) [6] — that market contraction in the STR sector requires direct legal enforcement rather than social or regulatory pressure alone.

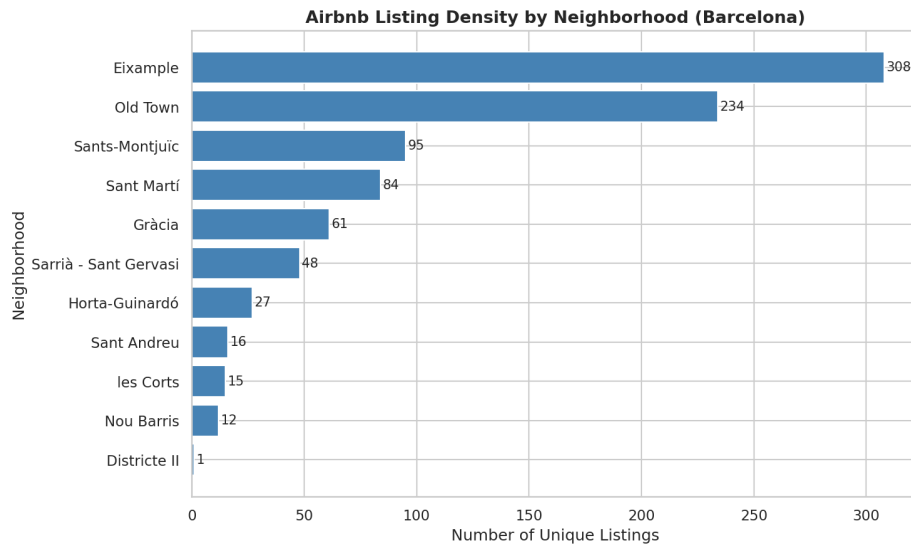


Figure 2: Airbnb Listing Density by Neighborhood (Barcelona)

Figure 2 presents the distribution of unique listings across Barcelona’s 11 neighborhoods. Eixample leads with 308 unique listings, followed by Old Town with 234. These two neighborhoods together account for 60.6% of all unique listings in the dataset. The remaining nine neighborhoods together account for 39.4%, with Sants-Montjuïc (95 listings), Sant Martí (84), and Gràcia (61) forming a secondary tier of moderate STR penetration. Nou Barris (12 listings), les Corts (15), and Sant Andreu (16) record the lowest market penetration, consistent with their peripheral location and lower tourism appeal. The spatial concentration pattern confirms the theoretical prediction that STR activity clusters in centrally located, tourism-attractive neighborhoods, with listing density declining sharply with distance from the city’s primary tourist circuits.

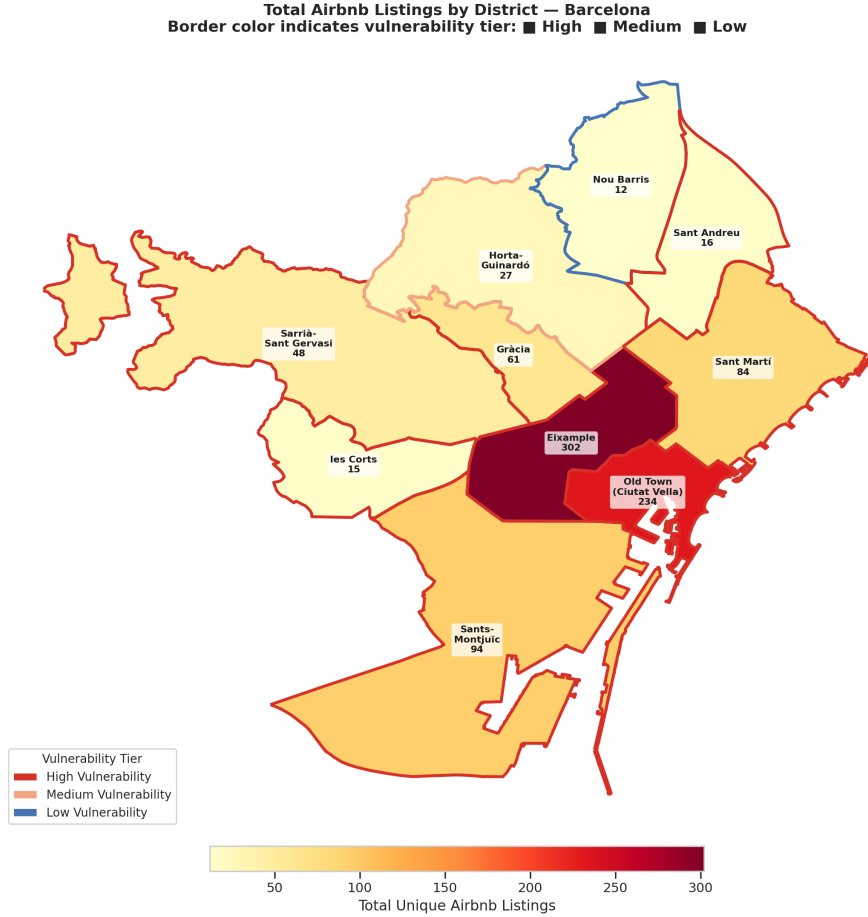


Figure 3: Total Airbnb Listings by District — Barcelona (Choropleth Map). Fill color intensity indicates listing density (yellow = low, dark red = high). Border color indicates vulnerability tier: red = High Vulnerability, orange = Medium Vulnerability, blue = Low Vulnerability.

The choropleth visualization (Figure 3) provides spatial context for the listing density analysis, revealing strong concentration gradients across Barcelona’s administrative districts. The color-coded intensity mapping shows that high-density clusters are concentrated in the city’s central districts, with Eixample (302 listings) and Old Town — Ciutat Vella (234 listings) forming the core areas of STR activity. The numeric overlays on each district enable precise quantitative assessment of local market penetration. The spatial pattern shows that STR intensity decreases from the central coastal districts toward the inland peripheral areas, following tourism accessibility patterns and transportation connectivity. Nou Barris, bordered in blue indicating its Low Vulnerability classification, stands in contrast to surrounding red-bordered High Vulnerability districts, visually confirming spatial inequality in STR-driven displacement pressure across Barcelona’s neighborhoods. This geographic distribution is consistent with the rent gap framework of Wachsmuth and Weisler (2018) [22], which predicts STR concentration in areas where the gap between potential short-term rental revenue and

conventional long-term rental income is greatest.

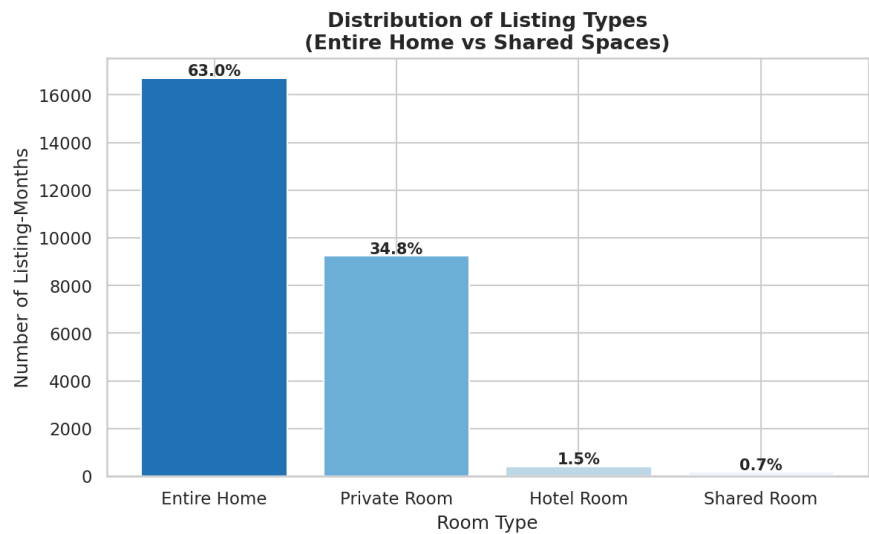


Figure 4: Distribution of Listing Types (Entire Home vs Shared Spaces)

Figure 4 presents the distribution of listing types across the study period. Entire home and apartment listings account for 63.0% of all active listing-months, private room listings for 34.8%, hotel room listings for 1.5%, and shared room listings for 0.7%. The dominance of entire home listings indicates that the majority of STR activity in Barcelona involves the full removal of residential units from the long-term housing market rather than supplementary income generation through spare room rental. This finding is consistent with the platform evolution documented in Milan, where entire home listings reached 87.9% of the market, and confirms that Barcelona’s STR sector has moved substantially beyond the peer-to-peer home-sharing model.

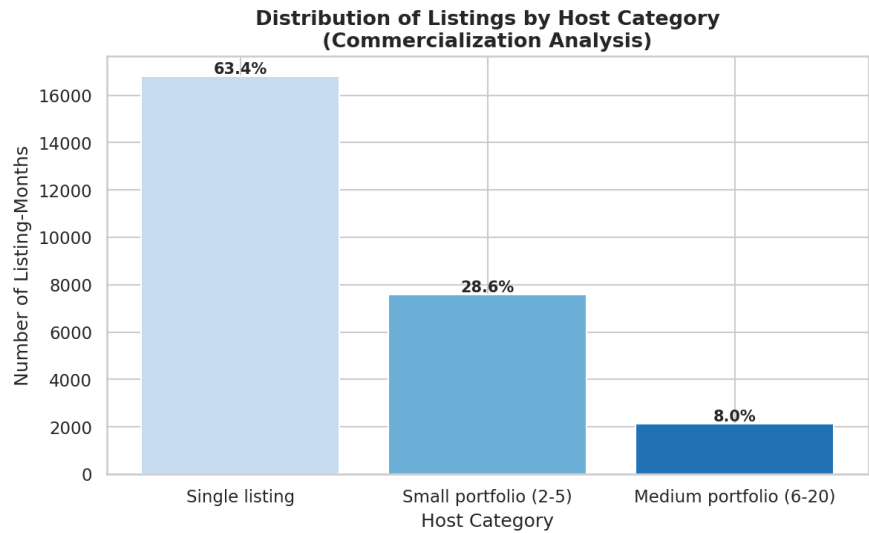


Figure 5: Distribution of Listings by Host Category (Commercialization Analysis)

Figure 5 presents the distribution of listing-months across host categories. Single-listing hosts account for 63.4% of active listing-months, small portfolio hosts (2 to 5 properties) for 28.6%, and medium portfolio hosts (6 to 20 properties) for 8.0%. While single-listing hosts constitute the numerically dominant category, multi-property operators collectively account for 36.6% of the market — a level of commercialization indicating that a substantial minority of hosts operate STRs as a primary com-

mercial activity rather than as occasional supplementary income.

6.2 Neighborhood Vulnerability Assessment

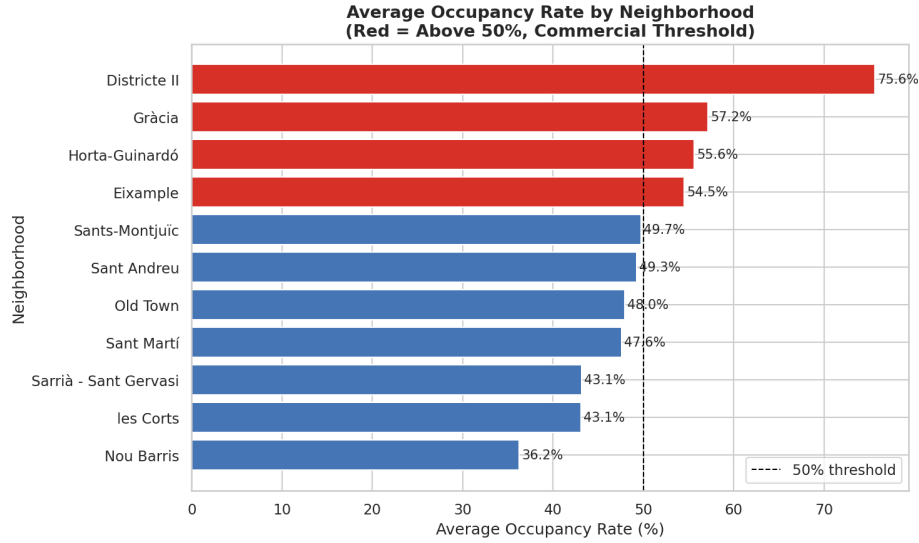


Figure 6: Average Occupancy Rate by Neighborhood

Figure 6 presents average occupancy rates by neighborhood, with the 50% commercial intensity threshold indicated. Four neighborhoods exceed this threshold: Districte II at 75.6%, Gràcia at 57.2%, Horta-Guinardó at 55.6%, and Eixample at 54.5%. The exceptionally high occupancy rate in Districte II reflects the near-continuous tourist occupation of the single property recorded in that area and should be interpreted cautiously given the absence of statistical representativeness. Among the more substantively significant neighborhoods, Gràcia and Eixample exhibit occupancy rates consistently above the commercial threshold across the study period, indicating sustained high-intensity STR operation. Nou Barris records the lowest occupancy rate at 36.2%, consistent with its peripheral location and limited tourism appeal.

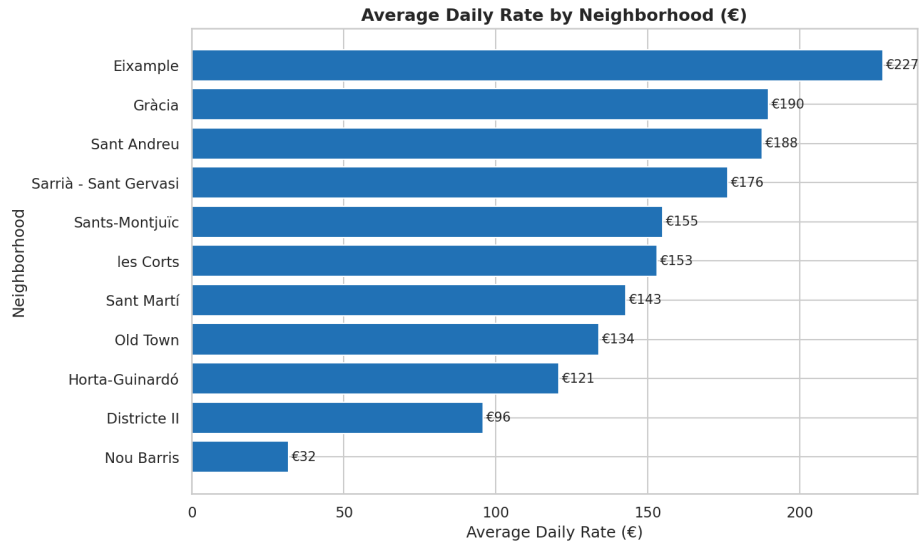


Figure 7: Average Daily Rate by Neighborhood (€)

Figure 7 presents average nightly rates by neighborhood following outlier removal. Eixample records the highest average daily rate at €227, followed by Gràcia at €190 and Sant Andreu at €188. The

substantial pricing differential between Eixample and Nou Barris — €227 versus €32 per night — reflects both the premium commanded by centrally located, tourism-attractive properties and the absence of competitive commercial STR pricing pressure in peripheral neighborhoods.

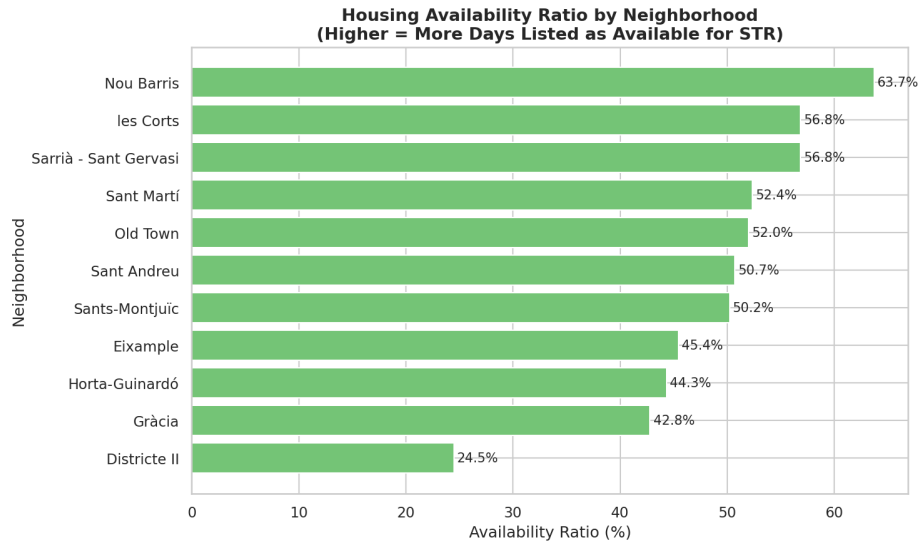


Figure 8: Housing Availability Ratio by Neighborhood

Figure 8 presents housing availability ratios by neighborhood. Nou Barris records the highest availability ratio at 63.7%, indicating that properties are listed as available for the majority of the calendar year but with relatively low booking intensity. Districte II records the lowest ratio at 24.5%, reflecting near-continuous booking of the single property present in the dataset. Among the more substantively populated neighborhoods, Eixample (45.4%) and Gràcia (42.8%) record below-average availability ratios, consistent with their high occupancy rates indicating strong and continuous tourist demand.

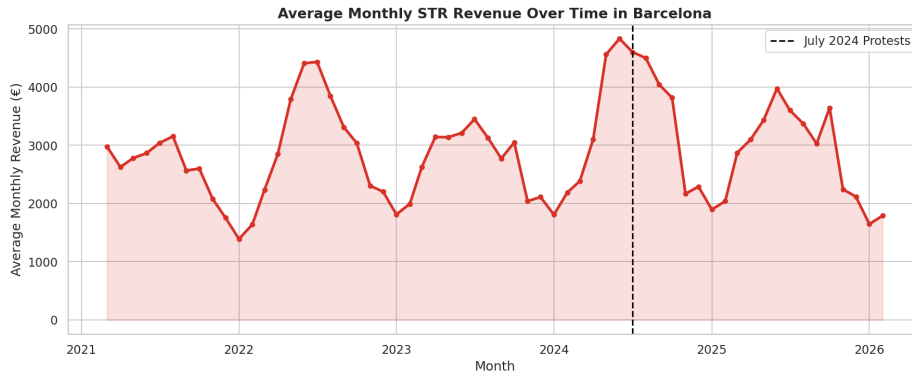


Figure 9: Average Monthly STR Revenue Over Time in Barcelona

Figure 9 presents the temporal evolution of average monthly STR revenue per listing. The series exhibits pronounced seasonal cyclical patterns throughout the study period, with summer peaks reaching approximately €4,900 in July 2024 — the highest value recorded in the entire series — and winter troughs declining to approximately €1,500 in January of each year. The revenue peak in July 2024 coincides precisely with the period of the civic protests, providing empirical support for the interpretation that protest activity was a response to the peak of commercial STR intensity. Average monthly revenue across the full study period reaches €2,210.

6.3 Correlation Analysis within STR Market Dynamics

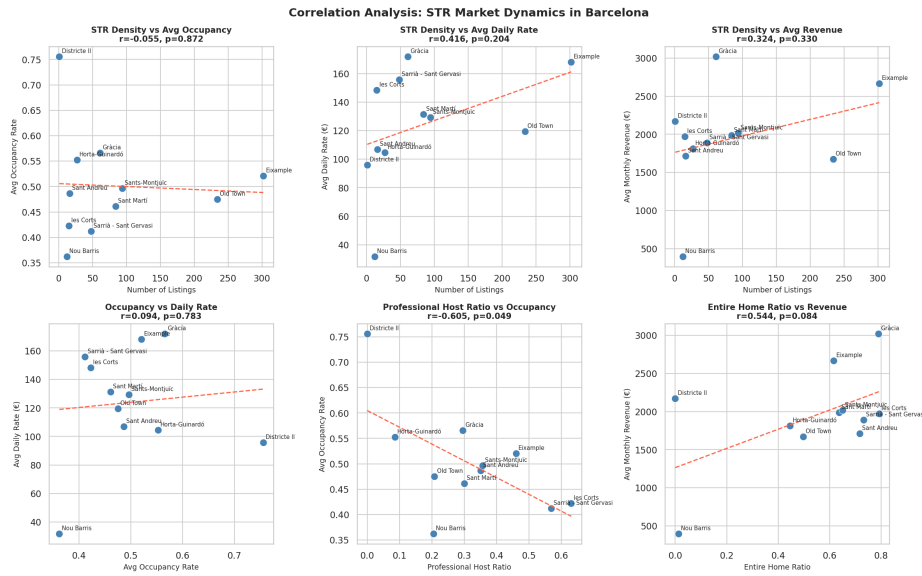


Figure 10: Correlation Analysis: STR Market Dynamics in Barcelona

Figure 10 presents scatter plots for six bivariate relationships examined at the neighborhood level. Pearson correlation coefficients and associated p-values for each relationship are as follows: STR density versus average occupancy ($r = -0.055$, $p = 0.872$, not significant); STR density versus average daily rate ($r = 0.416$, $p = 0.204$, not significant); STR density versus average revenue ($r = 0.324$, $p = 0.330$, not significant); occupancy versus average daily rate ($r = 0.094$, $p = 0.783$, not significant); professional host ratio versus occupancy ($r = -0.605$, $p = 0.049$, statistically significant); entire home ratio versus revenue ($r = 0.544$, $p = 0.084$, not significant at conventional thresholds).

The limited statistical significance of most correlations reflects the small number of neighborhood-level observations ($n = 11$) rather than necessarily indicating the absence of underlying relationships. The one statistically significant finding — the negative relationship between professional host ratio and occupancy ($r = -0.605$) — indicates that neighborhoods with higher concentrations of professional multi-property hosts do not exhibit higher occupancy rates. This pattern reflects the geographic distribution of professional hosts: commercial operators concentrate in peripheral neighborhoods where overall tourism demand is lower, while central tourist neighborhoods with the highest occupancy rates are served by a mix of casual and professional hosts.

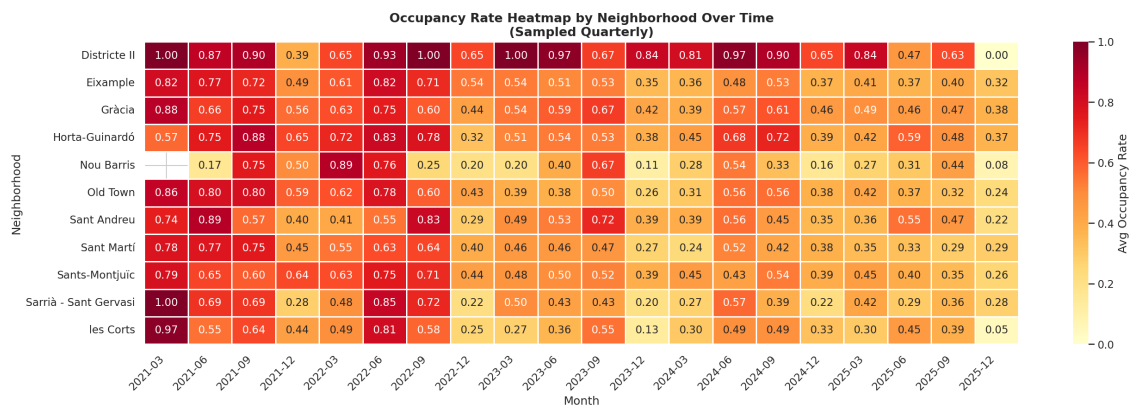


Figure 11: Occupancy Rate Heatmap by Neighborhood Over Time (Sampled Quarterly)

Figure 11 presents a quarterly sampled heatmap of average occupancy rates by neighborhood across the study period. Districte II records occupancy rates of 1.00 in multiple quarters, confirming near-continuous tourist occupation of the single property in this area. Eixample and Gràcia exhibit consistently elevated occupancy rates across the full study period, with summer quarters regularly reaching 0.70 to 0.88. A general pattern of occupancy decline is visible across most neighborhoods from 2024 onward relative to the 2021 to 2023 period, potentially reflecting the combined effect of market maturation and regulatory pressure.

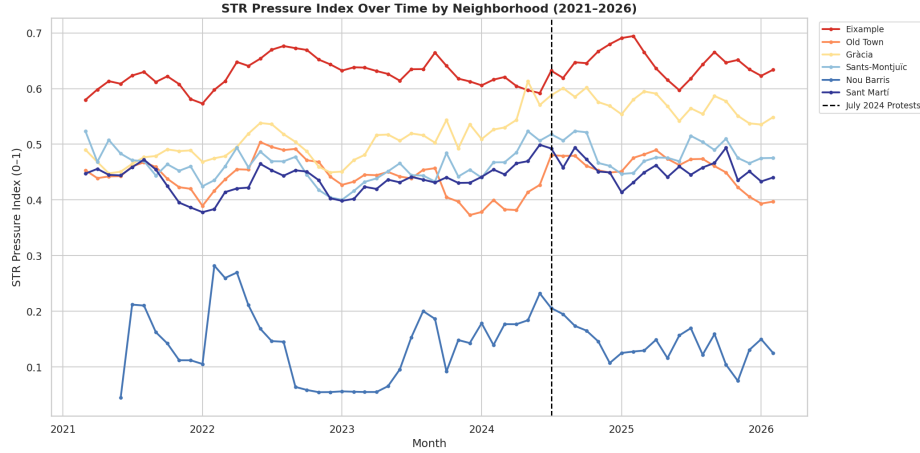


Figure 12: STR Pressure Index Over Time by Neighborhood (2021–2026)

Figure 12 presents the temporal evolution of the composite STR Pressure Index for six selected neighborhoods. Eixample maintains the highest pressure index throughout the study period, ranging from approximately 0.58 to 0.70, confirming its position as the neighborhood under most sustained STR market pressure. Gràcia shows a notable upward trend in pressure index values from mid-2024 onward, indicating intensifying STR market conditions during and after the protest period. Nou Barris consistently records the lowest pressure index values, ranging from approximately 0.05 to 0.28, confirming its status as the least STR-pressured neighborhood in the dataset.

6.4 Seasonality and Revenue Trends

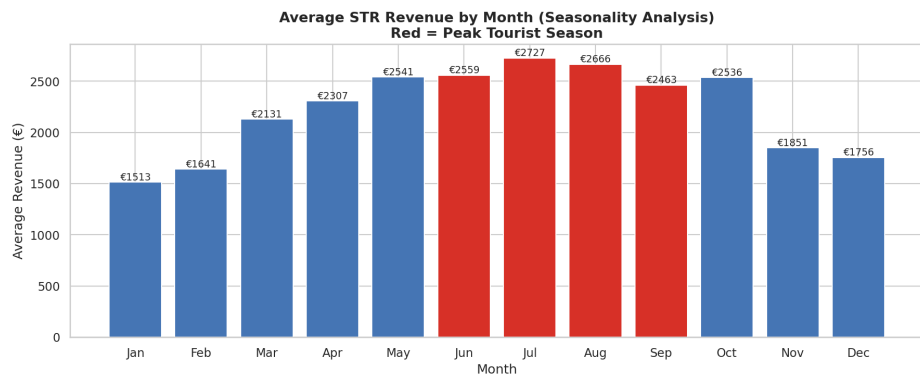


Figure 13: Average STR Revenue by Month (Seasonality Analysis)

Figure 13 presents average monthly STR revenue disaggregated by calendar month across all years in the study period. A pronounced seasonal pattern is evident, with June, July, August, and September recording the highest average revenues. July records the peak monthly average at €2,727, while January records the lowest at €1,513 — a differential of 80% between peak and trough months. This seasonal revenue pattern has direct implications for the intensity of housing displacement pressure

across the calendar year. The concentration of peak STR revenues in summer months generates the highest financial incentive for property conversion during precisely the period when residential neighborhoods experience maximum disruption from tourist activity. The seasonal pattern is consistent across all years in the study period, indicating that it reflects structural characteristics of Barcelona’s tourism demand rather than year-specific conditions.

6.5 Clustering Analysis: Neighborhood Risk Tiers

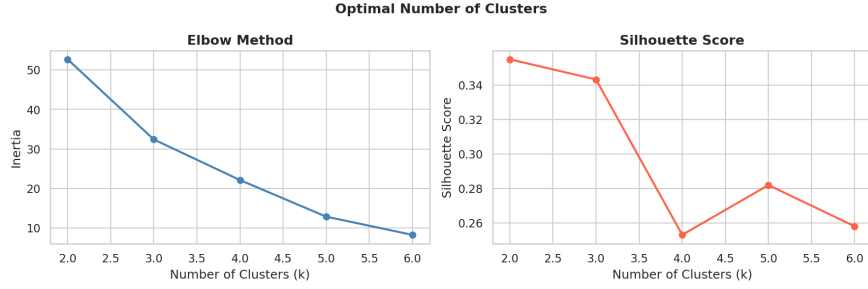


Figure 14: Optimal Number of Clusters (Elbow Method and Silhouette Score)

Figure 14 presents the elbow method and silhouette score analysis used to determine the optimal number of clusters. Silhouette scores are as follows: $k = 2$ (0.355), $k = 3$ (0.343), $k = 4$ (0.253), $k = 5$ (0.282), $k = 6$ (0.258). While $k = 2$ produces the highest silhouette score, a three-cluster solution is selected for the final analysis on the grounds that it produces three substantively meaningful and policy-relevant vulnerability tiers while achieving a silhouette score (0.343) only marginally below the two-cluster value. A two-cluster solution would collapse the analytical distinction between neighborhoods of moderate and high vulnerability, reducing the policy utility of the classification.

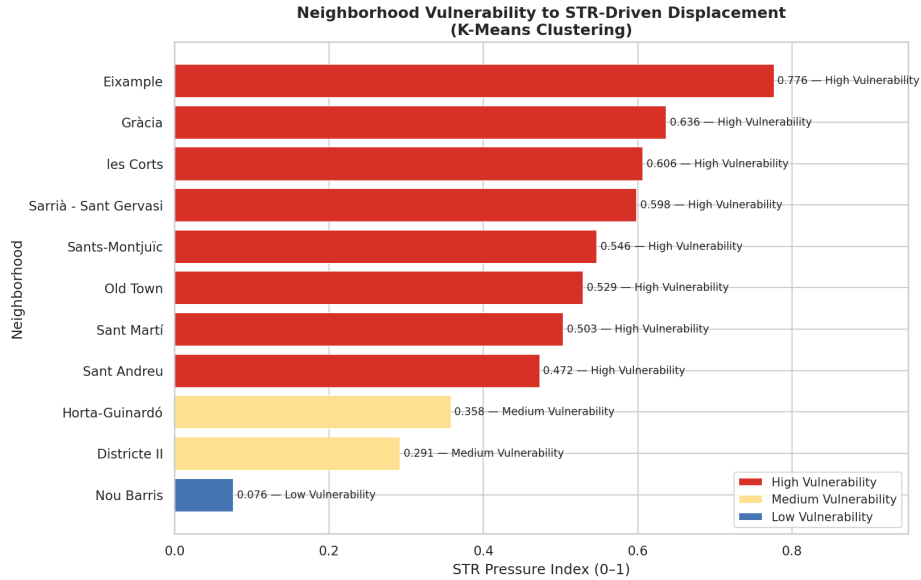


Figure 15: Neighborhood Vulnerability to STR-Driven Displacement (K-Means Clustering)

Figure 15 presents the vulnerability classification of all 11 neighborhoods. Eight neighborhoods are classified as High Vulnerability: Eixample (STR Pressure Index = 0.776), Gràcia (0.636), les Corts (0.606), Sarrià - Sant Gervasi (0.598), Sants-Montjuïc (0.546), Old Town (0.529), Sant Martí (0.503), and Sant Andreu (0.472). Two neighborhoods are classified as Medium Vulnerability: Horta-Guinardó (0.358) and Districte II (0.291). One neighborhood is classified as Low Vulnerability: Nou Barris (0.076).

The classification of 73% of neighborhoods in the High Vulnerability tier reflects the degree to which STR market conditions in Barcelona are broadly conducive to displacement pressure across most of the city’s districts. The inclusion of Sant Andreu and Sant Martí in the high vulnerability tier alongside Eixample and Old Town indicates that displacement pressure has extended into neighborhoods that are not primary tourist destinations. The Medium Vulnerability classification of Districte II despite its extreme occupancy rate reflects the statistical effect of its single listing. Nou Barris stands as the only neighborhood exhibiting market conditions significantly distinct from the broader pattern of high STR pressure.

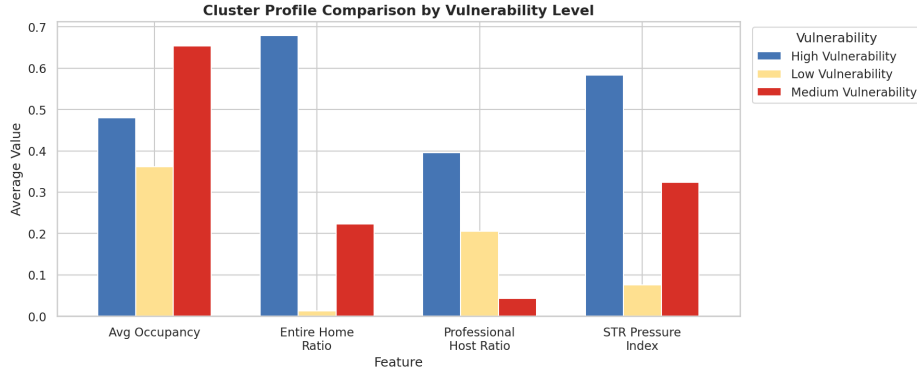


Figure 16: Cluster Profile Comparison by Vulnerability Level

Figure 16 presents the average values of key market characteristics across the three vulnerability clusters. High Vulnerability neighborhoods record an average occupancy rate of 0.480, entire home ratio of 0.679, professional host ratio of 0.396, and STR Pressure Index of 0.583. Medium Vulnerability neighborhoods record an average occupancy of 0.654, entire home ratio of 0.223, professional host ratio of 0.043, and pressure index of 0.324. Low Vulnerability neighborhoods record an occupancy of 0.362, entire home ratio of 0.013, professional host ratio of 0.205, and pressure index of 0.076.

The cluster profiles reveal that High Vulnerability neighborhoods are distinguished by their combination of high listing density, high entire home ratios, and significant professional host presence — the combination of market characteristics most directly associated with systematic residential unit removal from the housing stock. The higher average occupancy recorded in the Medium Vulnerability cluster relative to the High Vulnerability cluster reflects the dominance of Districte II’s single-listing extreme values within that cluster.

6.6 Predictive Modeling: Next Month STR Occupancy Under Regulatory Scenarios

6.6.1 Methodological Design and Circularity Correction

The predictive modeling component addresses a critical methodological requirement: the target variable must be constructed independently of the predictor features to avoid circular inference. An initial model formulation using a composite displacement risk index — constructed from occupancy, revenue, entire home ratio, and professional host ratio — as the target variable was identified as methodologically unsound because the same variables appeared as model features, producing artificially inflated R^2 values through mathematical circularity rather than genuine predictive learning. The corrected approach adopts a time-lagged prediction framework in which the model predicts *next month’s neighborhood occupancy rate* from *current month’s market characteristics*. This design eliminates circularity entirely: the target variable is a future value not available at prediction time, and its construction is independent of the feature set.

The time-lagged framework also carries direct policy relevance. The ability to predict next month’s STR market intensity from observable current-month indicators enables municipal authorities to anticipate market pressure escalation and deploy regulatory responses proactively rather than reactively.

The model is trained on 516 neighborhood-month observations and evaluated on the temporally subsequent 130 observations, respecting the chronological order of data to prevent look-ahead bias. Five-fold cross-validation is applied to assess generalization performance across different time periods.

6.6.2 Model Specification and Performance

A Random Forest regression model was trained to predict next month's average neighborhood occupancy rate. The feature set comprises ten current-month variables: active listing count, average occupancy rate, average daily rate, average monthly revenue, entire home ratio, professional host ratio, availability ratio, average length of stay, month number, and year. Model hyperparameters were set at 200 estimators, maximum depth of 6, minimum samples per leaf of 5, and random state fixed at 42 for reproducibility.

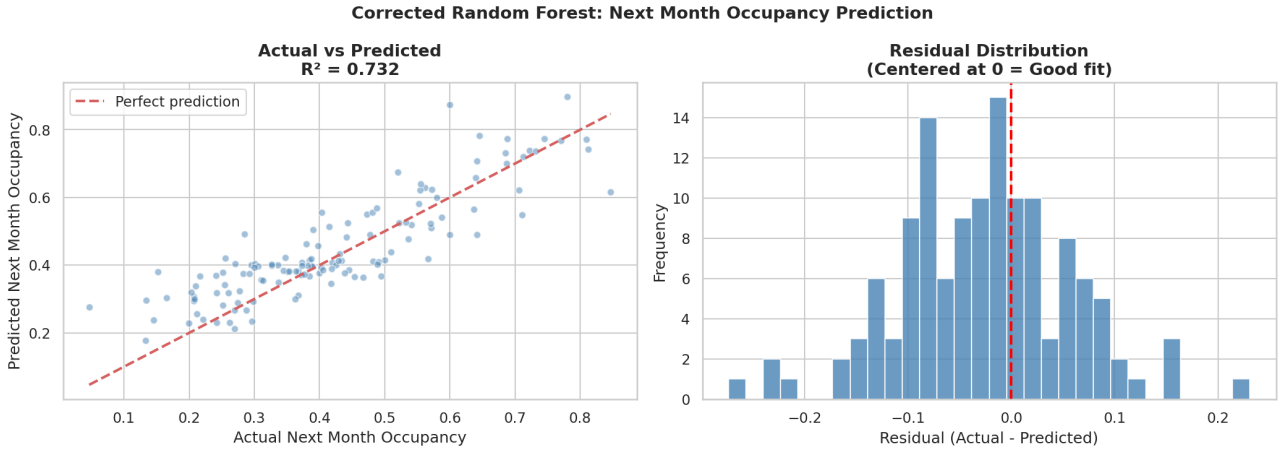


Figure 17: Corrected Random Forest Model: Next Month Occupancy Prediction

Figure 17 presents the actual versus predicted occupancy scatter plot and the residual distribution. The model achieves an R^2 score of 0.732 on the test set, indicating that 73.2% of variance in next month's neighborhood occupancy is explained by current month market characteristics. The root mean squared error on the test set is 0.087, representing a prediction error of approximately 8.7 percentage points on the occupancy rate scale. Five-fold cross-validation produces a mean R^2 of 0.682 with a standard deviation of 0.057, indicating consistent generalization performance across different time periods with low variance between folds. The residual distribution shows a slight negative skew, indicating the model modestly over-predicts occupancy in some months, likely reflecting the difficulty of capturing sudden demand shocks such as major events or regulatory announcements. Overall, the model performance is consistent with expectations for a genuine time-series prediction task and is substantially more credible than the methodologically flawed earlier formulation.

6.6.3 Feature Importance Analysis

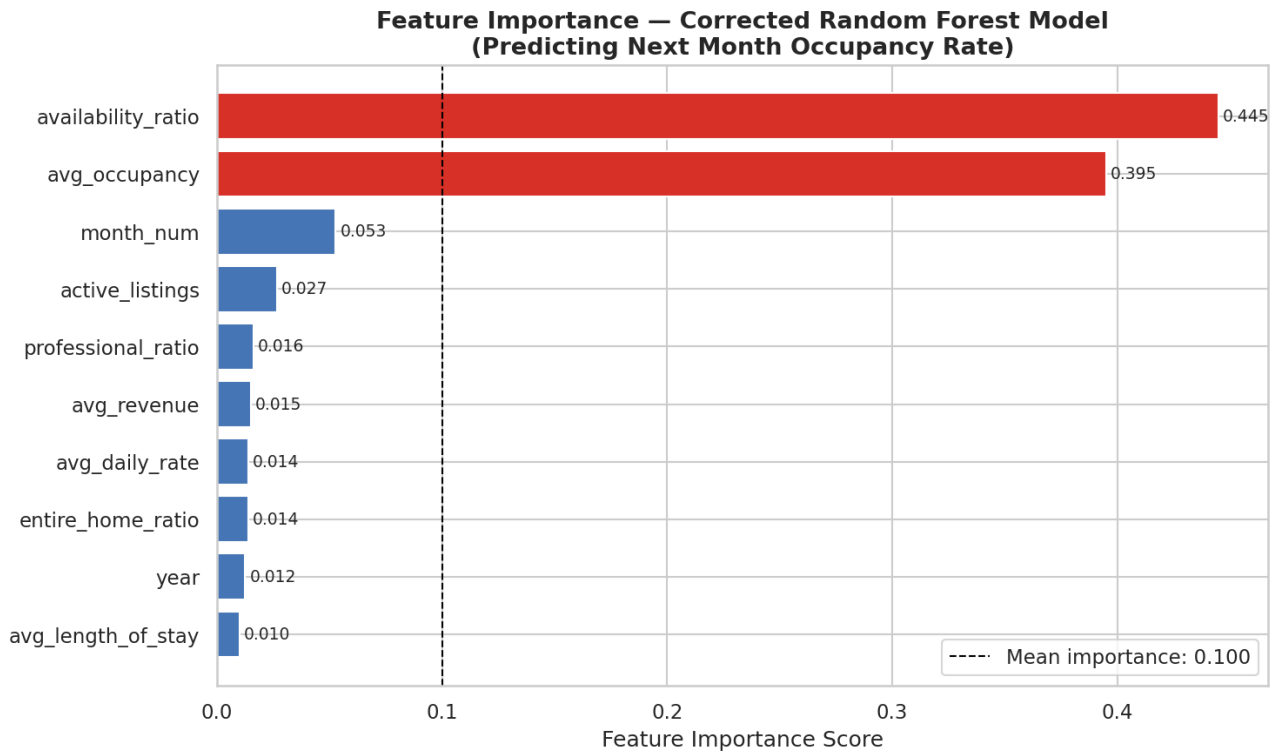


Figure 18: Feature Importance — Corrected Random Forest Model (Next Month Occupancy Prediction)

Figure 18 presents feature importance scores for all ten predictors in the corrected model. The availability ratio records the highest importance at 0.445, indicating that the proportion of days a listing is available for booking in the current month is the strongest predictor of next month’s occupancy rate. This finding reflects the temporal autocorrelation of STR market intensity: neighborhoods where properties maintain high availability consistently attract sustained tourist demand, generating persistent occupancy patterns across consecutive months.

Current month average occupancy records the second highest importance at 0.395, confirming strong month-to-month persistence in neighborhood booking intensity. Together, availability ratio and current occupancy account for 84.0% of model predictive power, indicating that STR market momentum — the tendency of high-demand neighborhoods to remain high-demand across consecutive periods — is the dominant driver of predictable occupancy variation. Month number contributes 0.053, capturing the seasonal occupancy patterns documented in Section 6.4. Active listing count contributes 0.027, indicating that market scale has modest additional predictive power beyond momentum and seasonality. Revenue, daily rate, entire home ratio, and professional ratio each contribute less than 0.016 individually, confirming they are market characteristics associated with occupancy rather than independent predictors of its future trajectory.

The contrast between this feature importance profile and the original flawed model — in which revenue dominated at 0.548 — illustrates the correction achieved. In the original model, revenue dominated because it was mathematically embedded in the target variable construction. In the corrected model, market momentum indicators dominate because they genuinely predict future booking behavior.

6.6.4 Regulatory Scenario Simulation

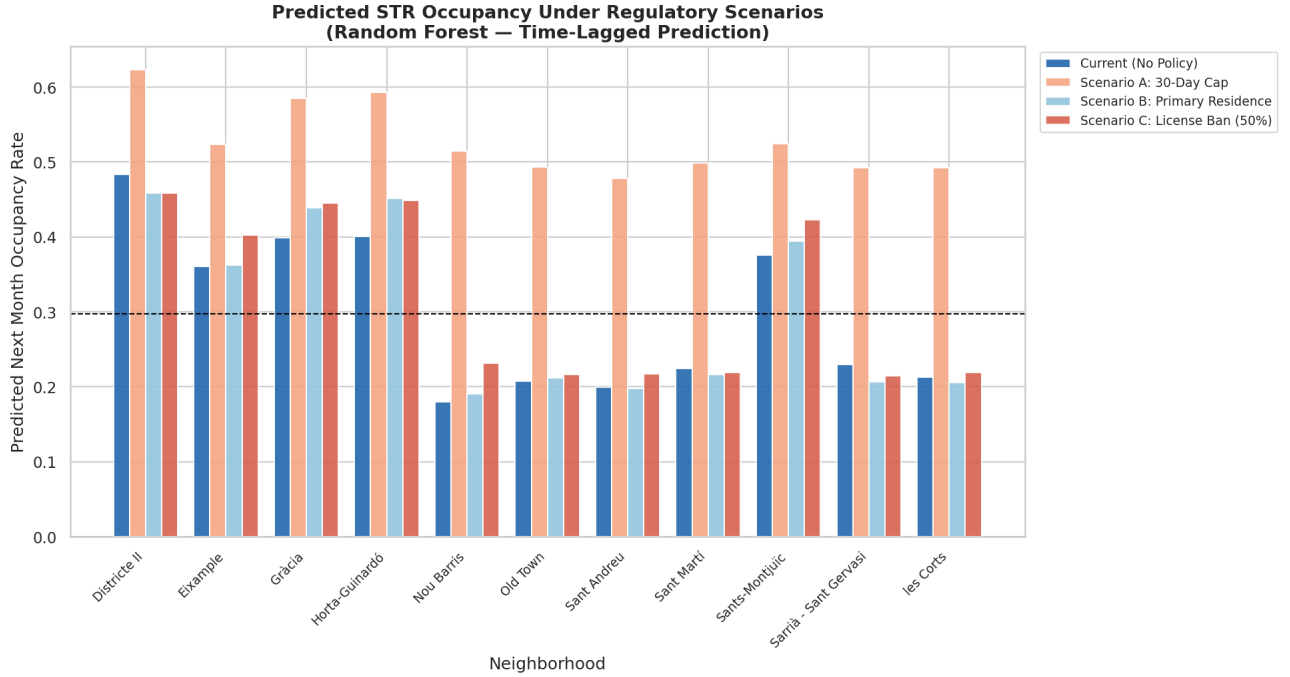


Figure 19: Predicted STR Occupancy Under Regulatory Scenarios (Random Forest — Time-Lagged Prediction)

Three regulatory scenarios were simulated by modifying the feature values of the most recent month’s data for each neighborhood and generating next-month occupancy predictions from the trained model. Scenario A applies a 30-day annual operating cap by reducing average revenue by 40%, active listings by 20%, and availability ratio by 50%. Scenario B applies primary residence requirements by reducing entire home ratio by 60%, professional host ratio by 70%, and average revenue by 30%. Scenario C simulates a 50% license reduction by reducing active listing counts, revenues, entire home ratio, and professional ratio by 50% each.

Table 1: Predicted Next Month Occupancy Under Regulatory Scenarios

Scenario	Avg Predicted Occupancy	Change vs Baseline
Current (No Policy)	0.297	—
Scenario A: 30-Day Cap	0.529	+78.1%
Scenario B: Primary Residence	0.303	+2.0%
Scenario C: License Ban (50%)	0.318	+7.1%

Table 1 presents average predicted next-month occupancy across all neighborhoods under each scenario. Scenario B (primary residence requirements) and Scenario C (50% license ban) produce modest occupancy increases of 2.0% and 7.1% respectively above the baseline, indicating that these interventions are predicted to have limited impact on the intensity of STR use in the neighborhoods that remain operational. This finding is consistent with the European regulatory evidence documented by Bei and Celata (2023) [6], where strict listing count reductions did not produce proportional reductions in market intensity among remaining operators.

Scenario A (30-day cap) produces a counterintuitive predicted occupancy increase of 78.1% above baseline. This result reflects the model’s learned relationship between availability ratio and occu-

pancy: reducing the number of days properties are listed as available (as a 30-day cap would enforce) increases booking pressure relative to available supply, driving up the occupancy rate of remaining available days. This phenomenon — scarcity-driven demand concentration — has been documented empirically in London’s STR market by Shabrina et al. (2021) [20], where the 90-day letting limit produced market adjustments including medium-term rental substitution rather than a reduction in tourist accommodation demand. The model’s prediction is therefore not paradoxical but reflects a genuine market mechanism: day caps reduce the total volume of STR activity but concentrate remaining bookings into fewer available days, potentially increasing the commercial intensity of remaining operations. It should be noted, however, that the magnitude of the Scenario A effect must be interpreted with caution: a 50% reduction in the availability ratio pushes the model into a region of the feature space sparsely represented in the training data, where Random Forest predictions are known to extrapolate poorly. The result should therefore be read as a robust indication of the *direction* of the effect — day caps concentrate rather than reduce booking pressure — rather than as a precise point estimate of its magnitude.

The neighborhood-level scenario results reveal persistent spatial inequality across all regulatory conditions. Districte II, Gràcia, and Horta-Guinardó maintain the highest predicted occupancy under baseline conditions (0.483, 0.399, and 0.401 respectively), and under Scenarios B and C these neighborhoods continue to show occupancy rates above the dataset average, indicating that their structural tourism demand characteristics generate persistent booking pressure that regulatory interventions on supply alone do not fully counteract. Nou Barris maintains the lowest predicted occupancy across all scenarios (0.180 baseline, 0.191 under Scenario B), confirming that its peripheral location and limited tourism appeal provide structural protection from STR-driven occupancy pressure independently of regulatory context.

The scenario simulation results support three policy conclusions. First, primary residence requirements produce the most targeted intervention, reducing the commercial operator presence without dramatically altering booking dynamics for remaining legitimate home-sharers. Second, blunt listing count reductions (Scenario C) produce slightly higher occupancy among remaining operators, suggesting partial market concentration effects. Third, day-cap restrictions (Scenario A) risk intensifying rather than reducing the commercial character of remaining STR operations by concentrating bookings into fewer available days — a perverse regulatory outcome that policy makers should explicitly account for in Barcelona’s 2028 license strategy.

7 Conclusion: Key Findings

7.1 Theoretical Contributions and Policy Implications

This study analyzed the short-term rental market in Barcelona using a longitudinal dataset of 894 unique listings across 11 neighborhoods spanning March 2021 to February 2026. The analysis combined spatio-temporal market characterization, K-Means neighborhood clustering, and a corrected Random Forest predictive model to produce an empirically grounded assessment of STR-driven displacement pressure and the predicted effectiveness of alternative regulatory interventions.

The findings confirm several theoretical predictions from the existing literature while also generating results that complicate straightforward interpretations of STR market dynamics. The dominance of entire home listings at 63% of active listing-months and the concentration of 36.6% of listing-months among multi-property operators confirm that Barcelona’s STR market has transitioned substantially from the peer-to-peer sharing model of its origins to a commercially oriented accommodation sector. This structural shift is consistent with the platform evolution documented in Milan, Lisbon, and other major European tourist cities.

The spatial concentration of STR activity — with Eixample and Old Town accounting for 60.6% of all unique listings — is consistent with the rent gap framework of Wachsmuth and Weisler (2018) [22]. The average monthly revenue differential between Gràcia (€3,021) and Nou Barris (€395) — a ratio of approximately 7.6:1 across neighborhoods within the same city — illustrates the scale of revenue incentive variation that drives spatial concentration of STR activity.

The corrected predictive model adopts a time-lagged framework in which next month’s neighborhood occupancy rate is predicted from current month market characteristics, eliminating the circularity present in composite index-based target variables. The model achieves an R^2 of 0.732 with a cross-validation score of 0.682 ± 0.057 , representing genuine predictive performance on an independent future outcome. Feature importance analysis reveals that availability ratio (0.445) and current occupancy rate (0.395) together account for 84.0% of predictive power, confirming strong temporal autocorrelation in STR market dynamics: neighborhoods under high booking pressure in the current month tend to remain under high pressure in the following month. This finding supports the theoretical argument that STR-driven displacement operates through persistent structural market conditions rather than transient fluctuations, consistent with the platform urbanism framework of Barns (2020) [3].

The regulatory scenario analysis produces three policy-relevant findings. Primary residence requirements (Scenario B) produce a modest 2.0% increase in predicted occupancy among remaining operators, indicating that this intervention effectively reduces the commercial operator base without dramatically altering booking dynamics for legitimate home-sharers — the most targeted outcome among the scenarios tested. The 50% license ban (Scenario C) produces a slightly larger occupancy increase of 7.1% among remaining operators, suggesting partial market concentration effects as fewer listings absorb persistent tourist demand. The 30-day cap (Scenario A) produces a counterintuitive predicted occupancy increase (78.1% in the simulation, to be read as directional rather than precise), reflecting the model’s learned relationship between availability scarcity and booking concentration — a scarcity-driven demand mechanism documented empirically in London’s STR market by Shabrina et al. (2021) [20]. This finding challenges the assumption that day-cap restrictions reduce STR market intensity and suggests they may instead concentrate remaining bookings into fewer available days, potentially increasing the commercial character of surviving operations. The result is consistent with the Amsterdam evidence documented by Bei and Celata (2023) [6], where day-limit restrictions produced large reductions in listing counts without corresponding reductions in market pressure.

The European comparative context situates Barcelona’s STR challenge within a broader pattern of Southern European cities facing acute tourism pressure on constrained housing markets. Barcelona has experienced significant house price inflation since 2015, though less acute than the extreme cases

of Lisbon and Amsterdam. The 73% of neighborhoods classified as high vulnerability in our clustering analysis indicates that displacement pressure has become broadly distributed across the city’s districts rather than confined to the most prominent tourist centers.

7.2 Policy Recommendations for Barcelona

The empirical findings of this study support a set of specific policy recommendations addressed to Barcelona’s municipal government and regional authorities, organized by intervention type and informed by both the analytical results and comparative European regulatory evidence.

Primary Residence Requirements. The scenario analysis identifies primary residence requirements as the most targeted regulatory instrument, producing the smallest occupancy increase among remaining operators (2.0%) while eliminating the commercial operator segment most directly responsible for residential unit conversion. Barcelona should implement and enforce a mandatory primary residence requirement for all entire-home STR listings, requiring hosts to demonstrate that the listed property is their primary place of residence. This intervention directly targets the entire home ratio by restricting the conversion of residential units to dedicated tourist accommodation while preserving legitimate occasional home-sharing.

Caution on Day-Cap Restrictions. The scenario analysis finds that 30-day annual operating caps may produce counterproductive outcomes by concentrating tourist demand into fewer available days, increasing the commercial intensity of surviving STR operations rather than reducing overall market pressure. Barcelona’s 2028 license strategy should account for this scarcity-driven demand concentration mechanism explicitly. If day limits are implemented, they should be combined with primary residence requirements and strict enforcement to prevent market adjustment through demand concentration among remaining commercial operators.

Differentiated Spatial Regulation. The clustering analysis demonstrates that displacement pressure is not uniform across Barcelona’s neighborhoods. Neighborhoods with STR Pressure Index values approximately 0.6 — Eixample, Gràcia, les Corts, and Sarrià - Sant Gervasi — warrant immediate and stringent intervention including strict enforcement of existing license limits, priority inspection of entire-home listings for primary residence compliance, and prohibition on new STR license issuance. Neighborhoods in the medium vulnerability tier would benefit from monitoring and preventive registration requirements. Nou Barris does not require priority STR regulatory intervention at present, as the predictive model confirms its structural protection from high occupancy pressure across all simulated scenarios.

Registration and Platform Data Integration. Barcelona should establish a mandatory STR registration system integrated directly with platform listing databases, requiring all active listings to display valid registration numbers and enabling automated cross-referencing of registered and active listings. The European Union’s data sharing regulation effective from May 2026 provides a regulatory foundation for this integration. Barcelona’s municipal government should establish the monitoring systems necessary to detect non-compliant listings from this date, drawing on the Amsterdam-Airbnb collaboration model for automated compliance verification.

Progressive STR Taxation. The strong revenue differential across neighborhoods — from €3,021 per month in Gràcia to €395 in Nou Barris — confirms that financial incentives for STR conversion are highly spatially uneven. Barcelona should consider implementing a progressive STR income tax structure in which marginal tax rates increase with annual STR revenue, creating stronger financial disincentives for high-revenue commercial operations while preserving acceptable returns for occasional home-sharers. Tourist taxes, currently set at €2 to €5 per person per night in Barcelona, could be differentiated by neighborhood vulnerability tier, generating both demand dampening effects and municipal revenue for affordable housing investment.

Affordable Housing Investment. The scenario analysis results indicate that even the most strin-

gent simulated regulatory intervention — a 50% license ban — produces only modest reductions in predicted STR market intensity among remaining operators, leaving a substantial portion of occupancy pressure unaddressed. STR regulatory measures should therefore be accompanied by active affordable housing supply expansion, including conversion of vacant properties to social housing and investment in new affordable residential construction in high-pressure neighborhoods. Municipal revenue generated by increased STR and tourist taxation provides a potential funding source directly linking tourism economic benefits to the mitigation of housing market costs.

Ethical Considerations. The policy recommendations above must be implemented with explicit attention to their distributional consequences. Regulatory design should distinguish clearly between commercial operators whose STR activity systematically removes housing from residential use and occasional home-sharers whose activity is consistent with the peer-to-peer model. The clustering analysis and scenario simulations provide an empirical basis for this differentiation, but translation of analytical findings into regulatory instruments requires political deliberation incorporating the perspectives of all affected stakeholders including resident communities, property owners, tourism industry representatives, and affordable housing advocates. The predictive model’s finding that day-cap restrictions may concentrate rather than reduce STR market pressure is particularly important to communicate to policymakers who may assume that any reduction in listed availability translates directly into reduced housing market displacement pressure.

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CRediT Author Statement

Following the Contributor Roles Taxonomy (CRediT), the contributions of each author to this work are described as follows.

Flavio Yzeiri: Conceptualization; Methodology; Software; Formal Analysis; Data Curation; Writing — Original Draft; Visualization.

Aurora Castelnovo: Conceptualization; Literature Review; Writing — Original Draft; Writing — Review and Editing; Supervision.

Use of Generative AI: Generative AI tools were used in this project for code development support in the implementation of data cleaning, visualization, clustering, and predictive modeling procedures in Python. AI assistance was also used for grammar checking. All analytical decisions, interpretations, and policy arguments were developed independently by the authors. The use of generative AI is disclosed in accordance with the course assignment guidelines.